Remote sensing of global change processes

Land cover and land use change:

Case study of forest extent change in Luxembourg from 2000 to 2020

by

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1. General introduction

Human activities (agriculture, logging, farming, urbanism…) are among the most widespread global land uses. In many countries, planning these land use practices is crucial and has led to significant environmental and socio-economic challenges. This has become evident in recent years with the increase in the amount of land allocated to agriculture, forestry and urban infrastructure. This expansion is closely linked to the growing demand by human beings (IPBES, 2019). Land Use change generally occurs through two main mechanisms: Land cover change (LCC) and Land use change (LUC). Understanding the distinction between them is fundamental to studying how human activities transform the Earth's surface. Land cover refers to the physical and biological elements present on the earth's surface, like vegetation, water bodies, urban structure… (Shunlin et al., 2012), and land use, on the other hand, refers to the function assigned by humans to the land and includes social, economic and cultural dimensions such as agriculture, forestry, conservation… (Shunlin et al., 2012). These two terms are often grouped in remote sensing, which primarily detects land cover from spectral signatures, but the analysis is made in the context of land use (Zhu et al., 2022).

Although humans have been modifying land to obtain food and other resources for many years, the current extent and intensity are greater now than in the past, leading to change at global and local scales (Winkler et al., 2021). Therefore, land use change is now considered a key driver of environmental change, with up to 80% of the Earth's surface being directly or indirectly affected (Almut, 2017). In many countries in the global North, forest cover has increased through reforestation or natural regeneration; in contrast, in the global South, the decline of forest area is remarkable and caused mainly by agricultural and built-up area expansion (Winkler, 2023). As a country in the global north, the context of land use change in Luxembourg is influenced by various factors, including urbanisation, natural hazards and climate change. Land use in Luxembourg is mainly dominated by agricultural areas and forests, while artificial surfaces, such as urban and industrial areas, occupy a smaller but increasing territory (Basse et al., 2016). Previous land-use studies based on Corine Land Cover data have shown that urban and industrial areas expanded between 1990 and 2006, while agricultural land slightly decreased over the same period. Forest areas remained relatively stable, indicating that Luxembourg has historically experienced moderate forest change compared with the stronger expansion of artificial land uses (Basse et al., 2016).

However, even limited changes in forest cover can be important in a small and densely connected country, particularly in relation to biodiversity conservation, landscape fragmentation, ecosystem services and sustainable spatial planning.

Remote sensing provides an effective basis for LULC analysis because it enables repeated observation of large areas of the Earth's surface over time. Through multi-temporal satellite imagery, it is possible to detect, map and quantify changes in land cover and land use with consistent spatio-temporal coverage (Winkler et al., 2021). This is particularly important for forest monitoring, where satellite data can support the identification of forest loss and assess vegetation dynamics. LULC analysis is organised according to a five-facet framework that examines where changes occur, when it occurs, what land cover or land use class is affected, how the change develops in terms of magnitude, direction, duration and rate (Zhu et al., 2022). Several remote sensing methods can support this analysis, including image classification followed by post-classification, which identifies transitions between land cover classes (Lu et al., 2004). Change vector analysis, which measures the magnitude and the direction of spectral change (Chen et al., 2003) and time series analysis, which detects abrupt or gradual changes in vegetation dynamics over time (Yang & Lo, 2002). Using the Landsat L3 dataset, this project aims to detect and quantify forest cover change in Luxembourg between 2000 and 2020, with particular focus on the spatial extent of the forest loss and the region most affected. This study develops a methodological approach based on Earth observation data to monitor and quantify changes in forest cover. The approach is applied to the detection and mapping of deforestation in Luxembourg using the Landsat-based Level-3 dataset for the period 2000-2020. Forest cover change is assessed through change detection analysis to quantify the extent and spatial distribution of forest loss over time. The study addresses two main research questions:

- What are the dominant land-cover classes in Luxembourg, and how did their mapped extent change between 2000 and 2020?
- What was the spatial extent and distribution of forest disturbance in Luxembourg between 2000 and 2020?

2. Methodology

2.1. Study area

This study focuses on the Grand Duchy of Luxembourg, a country in Western Europe with an area of 2586 square kilometres, bordered by Belgium (to the west and north), France (to the south and west) and Germany (to the east). It is located between 49°-51° latitudes north and 5°-7° longitudes east (Gerber et al., 2017).

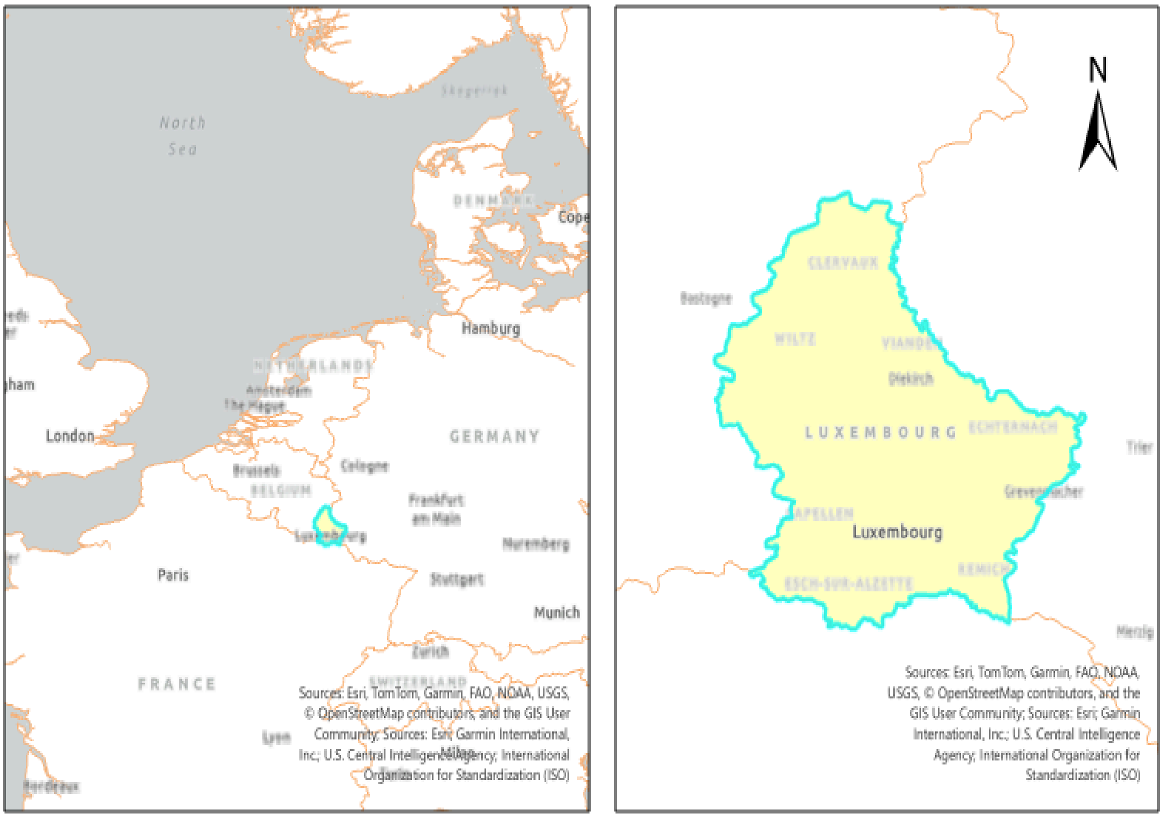

Figure 1: Overview of study area

From a physical perspective, Luxembourg is characterised by marked landscape diversity, which influences the spatial organisation of land use. The northern part of the country belongs to the Oesling, where the landscape is associated with the Ardennes and is generally more elevated and dissected by valleys with an average elevation of approximately 450 m (Slotboom and van Mourik, 2015). While the central and southern parts of the Gutland are characterised by alternating sedimentary formations with different resistance to erosion (Slotboom & Van Mourik, 2015). These topographic contrasts are important for spatial planning because relief, slope, and terrain fragmentation can influence construction, agricultural use and forest distribution. Basse et al. (2016) explicitly include slope gradient as one of the physical determinants used to model land use dynamics in Luxembourg (Watson et al., 2019).

The Luxembourg land use classification system is organised into three hierarchical levels. Level 1 describes broad land-cover categories such as Settlement, Agriculture, Forest, Natural Surfaces, Water and transport, while Level 2 and 3 provide progressively more detailed subclasses (space4environment, 2020). Forests constitute a major component of Luxembourg's landscape and cover approximately 92150 ha, corresponding to about 35% of the natural territory. Broadleaved species represent around 64% of this forest area, while coniferous species account for approximately 36% (The Government of the Grand Duchey of Luxembourg, 2026). The spatial distribution of forests is that around 55% of forested area is located in the northern region of the country, characterised by more pronounced relief. These forests also have an important social and recreational role, as more than 85% of the forest areas are located within 1500 m of built-up areas, making them highly accessible to the population (The Government of the Grand Duchey of Luxembourg, 2026).

2.2. Datasets

Over the past few decades, the use of Landsat-derived satellite imagery in forestry has grown due to the availability of a long time series of satellite observations of land cover and its dynamics (Gómez-Fernández et al., 2024). Due to its consistency in change detection, Landsat plays a crucial role in the systematic monitoring of Earth's surface (Hemati et al., 2021). The Global Land Cover and Land Use Change (GLCLU) Landsat-based dataset was utilised in this study. This dataset was produced by the Global Land Analysis and Discovery (GLAD) laboratory from the University of Maryland. The dataset provides globally consistent, annually updated land-cover classification at a 30-meter spatial resolution derived from Landsat data from 2000 to 2020. It includes a range of thematic land cover products, including cropland, forest height and extent, urban areas, surface water, snow and ice. The GLCLU dataset was accessed through the Google Earth Engine (GEE) platform, where data is available under: https://glad.earthengine.app/view/glcluc-2000-2020. These layers represent global land-cover classifications for the years 2000, 2010, and 2020. For this study, the layers were spatially subset to the national boundaries of Luxembourg.

For creating the GLCLU dataset, only snow-free season observations were used in temperate and boreal regions. The ARD (Analysis Ready Data) product consists of 16-day global composites of normalised surface reflectance and brightness temperature generated from Landsat 5, 7 and 8 Collection 1 Trier 1 archive. The processing includes Radiometric and atmospheric correction; snow, cloud and cloud shadow detection using a quality assessment layer and finally the aggregation of the best-quality observations into temporally consistent 16-day composites (Potapov et al., 2022).

To enable robust multitemporal classification, after excluding cloud and shadow and applying temporal gap-filling, some phenology metrics were calculated. The phenology metrics include Statistics (minimum, maximum and quartiles), average, amplitude, vegetation and water indices (NDVI & NDWI) and the brightness temperature Spectral reflectance percentiles.

To improve environmental representation, the spectral metrics were augmented with topographic variables (elevation and slope) derived from the Shuttle Radar Topography Mission (SRTM) and ASTER Global DEM datasets (Potapov et al., 2022).

The GLCLU dataset was produced using a class-specific modelling strategy, where each thematic land cover type was mapped using a classification approach adapted to its spectral, temporal and structural characteristics :

- **Cropland** was mapped using supervised classification based on bagged decision tree ensembles. Cropland presence was identified if the same crop area was detected during any year within the 2000-2003 and 2016-2019 period
- **Forest height and extent** were estimated using regression tree models calibrated with GEDI lidar RH95 metrics. Forest was defined as woody vegetation with a height >= 5 m at the Landsat pixel scale.
- **Built-up areas** consisted of all man-made infrastructure (buildings, roads…) and residential land uses. They were mapped using a deep learning convolutional neural network (U-Net architecture) trained on OpenStreetMap building and road data, complemented by manually interpreted samples. The model produced per-pixel probabilities that were thresholded to generate built-up land extent for 2000 and 2020.
- **Open surface water** was mapped using a screen-based classification of individual Landsat observations from 1999 to 2020, followed by a time series of annual water presence.

With an overall accuracy of 85%, the GLCU land cover products for 2000 and 2020 were used as input datasets for land cover mapping, area statistics computation and land cover change analysis over Luxembourg. In addition, a point dataset was created to validate the change map produced later in the analysis. The use of a harmonised, global dataset ensures methodological consistency and facilitates meaningful temporal comparison between the two reference years.

2.3. Method

The analysis was implemented using a combination of Google Earth Engine (GEE), QGIS and RStudio. GEE was used to access and download GLCU data subsetted to Luxembourg boundaries; QGIS for spatial post-processing, map production and visual inspection of change patterns; RStudio was used for summarising statistics, plotting and accuracy assessment.

2.3.1. Acquisition and preprocessing of GLCLU land cover map

The first step involved extracting GLCLU land-cover maps for the years 2000, 2010, and 2020 for the Luxembourg territory. This was done in GEE using the global GLCLU land-cover layers and a national boundary polygon for Luxembourg (Appendix 1). The resulting land-cover raster was exported at 30 m resolution for use in subsequent analysis and cartographic visualisation. The very small patches and isolated pixels were removed, and Classes with the same land cover type were aggregated into one. The Forest classes, for instance, were grouped into two classes: open forest with tree height between 5 and 25 m and dense forest with tree height > 25 m (Fig. 2).

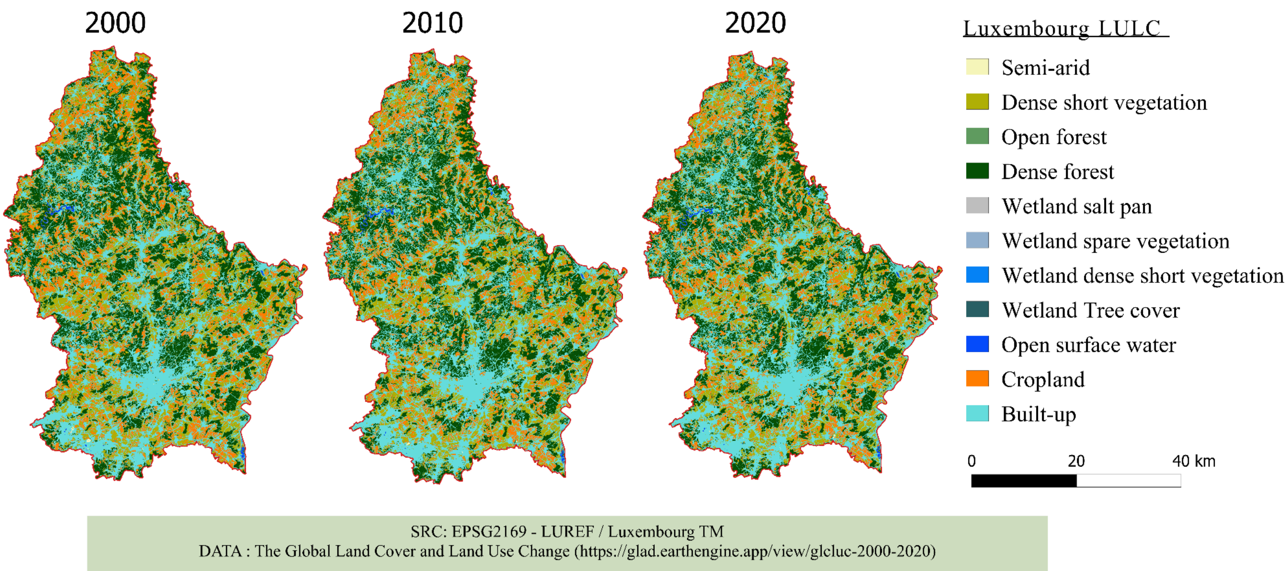

Figure 2: Land cover map

The extracted land-cover maps illustrate the spatial distribution of land cover classes across the study area. Forest dominates the landscape over the study period and represents the most extensive land cover type. Agricultural areas are concentrated around urban areas, and shrubland serves as a traditional vegetation zone between agricultural areas and dense forest cover. Urban areas appear as small but distinct clusters distributed along transportation corridors and among major settlements, while bare-soil areas are scattered throughout the region. To quantify the temporal changes in land cover, class-specific area statistics were calculated.

2.3.2. Land cover map processing

For the forest-focused analysis, the GLCLU were simplified into a binary forest / non-forest classification for each year (2000, 2010, 2020). Based on the GLCLU class legend, all classes corresponding to woody vegetation (forest types) were grouped into the forest category, while all remaining classes were grouped into non-forest. Forest pixels were assigned a value of 1. At the same time, all the other classes were grouped as non-forest (value 0). This simplification reduces thematic noise between non-forest subclasses and improves the post-classification comparison (Fig. 3). To see how much forest area Luxembourg has gained or lost over time, net change was also computed (Winkler, 2023).

$$\text{Net change} = FA_{2020} - FA_{2000}, \quad \textit{Where} \quad FA_t = \textit{the area of the forest in year } t$$

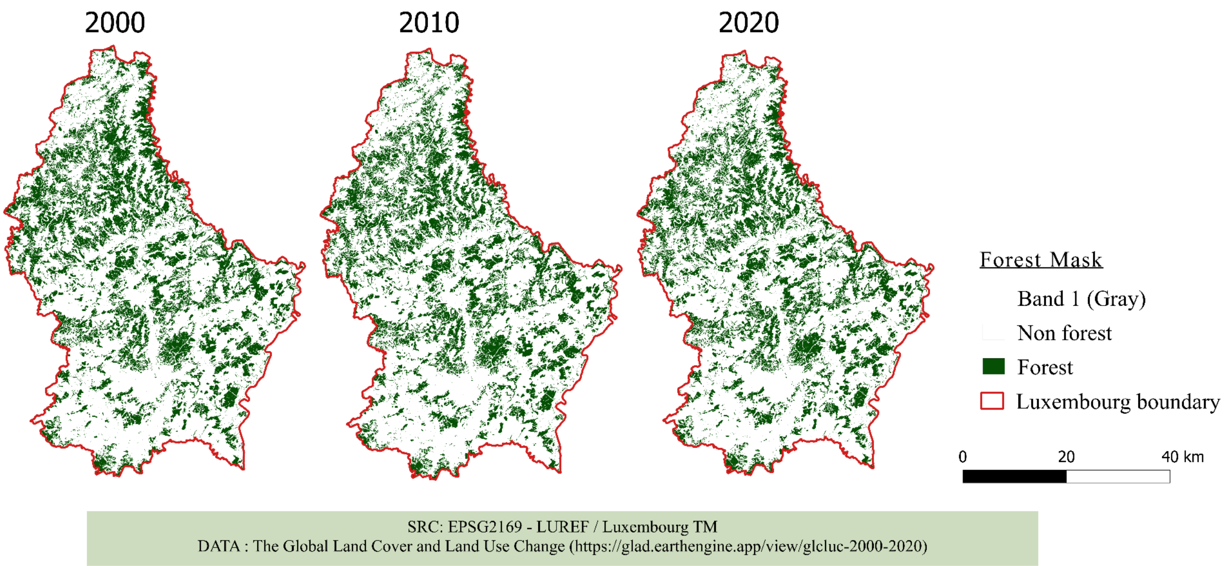

Figure 3: forest and non-forest map here

To capture not only net change but also the temporal trajectories of forest cover, the three binary forest masks were combined into a single trajectory map (change map). A unique change code was generated, representing the temporal trajectory of each pixel. A 3-digit code was computed as :

$$\text{Change code} = 100 \times F_{2000} + 10 \times F_{2010} + 1 \times F_{2020}$$

Where $F_t = 1$ *if the pixel was forest in year t and* $F_t = 0$ *otherwise*

The change map was computed, and for each trajectory class, the total area and percentage were calculated, providing detailed information on stable forest and non-stable forest, as well as forest loss and gain between 2000 and 2020.

Table 1: Change map interpretation logic

2000	2010	2020	Labels
1	0	0	Loss 2010
1	1	0	Loss 2020
1	1	1	Stable Forest
0	1	1	Gain 2010
0	0	1	Gain 2020
1	0	1	Unstable trajectory
0	1	0	Unstable trajectory
0	0	0	Stable non-forest

The temporary trajectory classes 101 and 010 represented only 0.26% and 0.04% of the study area, respectively (Tab. 2). Because these classes likely reflected classification noise or spectral inconsistencies, they were visually inspected using Google Earth time-series imagery. Pixels in class 101 were reclassified as stable forest when no persistent forest-loss event was visible, while pixels in class 010 were reclassified as stable non-forest when no persistent forest establishment was visible.

Table 2: Unstable classes

Code	Reclassified	Count	area (ha)	proportion (%)
101	stable Forest	12942	680.1667908	0.263018867
010	stable non-forest	1740	91.44569742	0.035361832

2.3.3. Accuracy assessment and validation

The reliability of the forest-change map was assessed using a stratified random sampling approach combined with an area-adjusted accuracy estimation following Olofsson et al.(2014). Before sampling, pixels labelled as unclassified were excluded from the change map classes. A stratified random sampling design was used. To ensure that rare change classes were represented, a minimum of 10 validation points was assigned to each rara class, while the remaining samples were allocated to the dominant stable classes according to their mapped area (Tab. 3).

Table 3: Validation points

	Loss 2010	Loss 2020	Stb. non-forest	Stb. forest	Gain 2010	Gain 2020	Total
Val. Points	10	10	39	17	10	10	96

This strategy ensures efficient representation of rare change classes while maintaining proportional sampling of dominant classes.

The validation dataset contains two labels for each sample location: the mapped class extracted from the change map and the reference class determined through visual interpretation of the Google Earth series imagery. A custom R script (Appendix) was used to construct the confusion matrix, derived from the area-proportion error matrix, and calculate area-adjusted accuracy metrics. The limitation of this method is the total sample size, which is relatively small, and therefore limits the precision of accuracy metrics and can lead to bigger confidence intervals in area-weighted estimates.

3. Results

3.1. Land use dynamics in Luxembourg between 2000 and 2020

The temporal dynamics of the main land-cover classes in Luxembourg between 2000, 2010 and 2020 are summarised in Figure 4. The figure shows the mapped area of each class for the three reference years. Overall, built-up areas/artificial surface, dense forest and dense short vegetation represent the largest mapped classes, while open forest and cropland occupy smaller proportions of the study area. Built-up areas increased over the study period, indicating continued expansion of artificial surfaces. In contrast, dense short vegetation and agricultural classes show comparatively smaller changes, while forest-related classes remain relatively stable, with only moderate variation between the three dates.

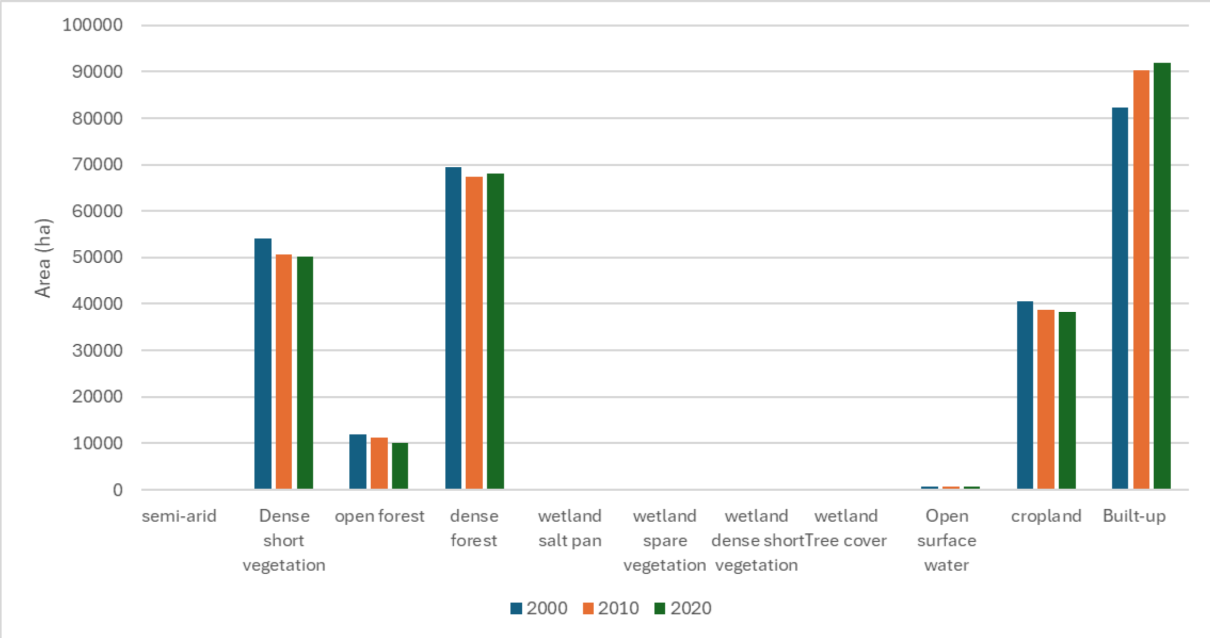

Figure 4: Land cover classes area statistics

The mapped area of the main GLCLU land-cover classes in Luxembourg for 2000, 2010 and 2020 remains stable across the three years, at approximately 259300 ha. Built-up area/artificial surfaces increased from 82297 ha in 2000 to 91874 ha in 2020, indicating an expansion of artificial land cover classes. In contrast, total forest area, calculated as the sum of open and dense forest, decreased steadily from 81406 ha in 2000 to 78085 ha in 2020. This corresponds to a net mapped forest decrease of approximately 3321 ha. Dense short vegetation and cropland also decreased slightly, while minor classes such as wetlands, semi-arid surfaces and open water bodies occupied only small areas and showed limited temporal variation.

3.2. Forest-cover dynamics **in Luxembourg between 2000 and 2020**

The spatial distribution of change derived from the GLCLU dataset reveals that stable forest dominates most of the study area, particularly in the northern and eastern parts of the country, where larger and continuous forest patches are visible (Fig. 5). In contrast, the stable non-forest class is much more prevalent in the central and southern parts of the country.

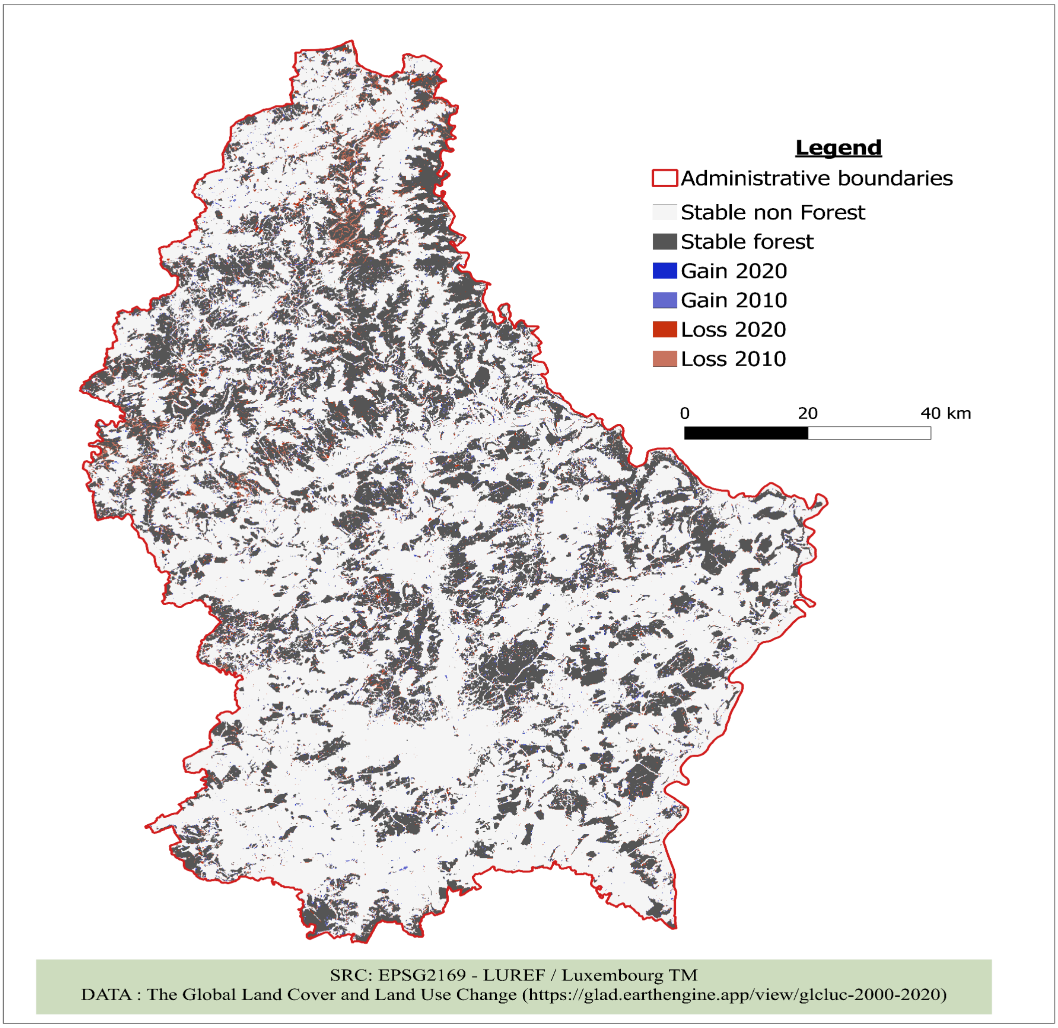

Figure 5: Forest change map

The map identifies six classes: stable forest, stable non-forest, forest loss in 2000, forest loss in 2020, forest gain in 2020 and forest gain in 2000. Loss of forest is not uniformly distributed but rather concentrated in specific zones, suggesting a localised disturbance process. Areas of loss appear mainly as scattered, with clusters visible along the margin of existing forest patches. Loss in 2020 seems more frequent and extensive than Loss in 2010. Forest gains are comparatively sparse and occur mainly in small, isolated patches, indicating limited regeneration. In addition, the gain in 2020 is more widespread than the one in 2010; forest expansion increased in the latter period, but the gain remains smaller and fragmented.

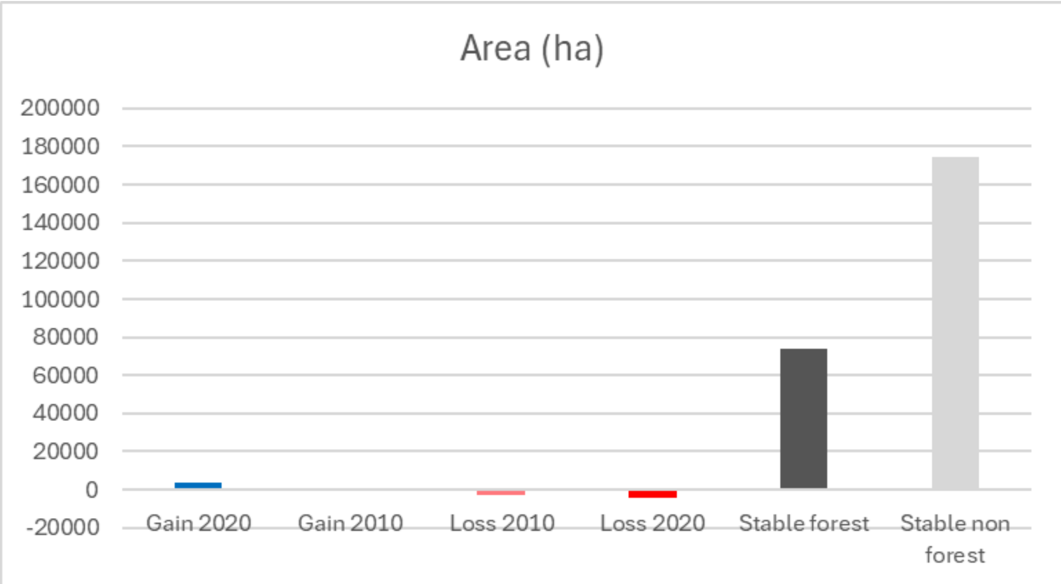

Figure 6: Diagram of forest change area

To quantify the magnitude of change, both mapped and area-adjusted estimates were analysed (Table 4). The mapped proportions indicate that forest loss accounts for 1% in 2010 and 1.7% in 2020 of the total study area. Combined, these results give a gross forest loss of approximately 2.7% over the study period. In contrast, forest gain remains very low with an overall value of 1.62 (with 0.22% in 2010 and 1.4 in 2020), indicating an insufficient recovery of forest loss over the study period.

Table 4: Table with forest change class metrics

	Mapped area (ha)	Mapped Proportions (%)	Adjusted Area (ha)	95% CI half-width(ha)
Loss 2010	2597.74	1.0018	6200.05	8838.88
Loss 2020	4333.42	1.6712	8077.83	8934.65
Stb. non-forest	174245.47	67.1965	157500.41	16844.37
Stb. forest	73802.04	28.4612	80532.35	14973
Gain 2020	3750.85	1.4465	1125.26	1122.99
Gain 2010	577.95	0.2229	5871.58	8609.21

The adjusted area estimates derived from the validation samples indicate that stable non-forest dominates the study area, with an adjusted area of 157500.41 ha, followed by stable-forest with 80532.35 ha. Forest loss was estimated at 6200.05 ha in 2010 and 8077.83ha in 2020, while forest gain was 5871.58 ha in 2010 and 1125.26 ha in 2020. Considering the adjusted estimates of forest loss and gain, the results indicate that total forest loss is about 14277.88 ha while total forest gain is 6996.84 ha, resulting in a net forest decrease of approximately 7281.04 ha between 2000 and 2020.

3.3 Accuracy assessment and validation

The accuracy assessment was used to evaluate the reliability of the trajectory-based forest-change map. In total, 96 validation points were interpreted using Google Earth imagery and compared with the mapped change classes. The resulting area-adjusted error matrix is presented in Table 5. This matrix accounts for the unequal area proportions of the mapped classes and provides the basis for the calculation of the user's accuracy (UA), producer's accuracy (PA), overall accuracy (OA) and adjusted area estimates.

Table 5: area-adjusted confusion matrix

	Loss 2010	**Loss 2020**	**Stb. non-forest**	**Stb. forest**	**Gain 2020**	**Gain 2010**
Loss 2010	0.005	0.001	0.002	0.0020	0	0
Loss 2020	0.0017	0.01	0.0017	0.0017	0	0.0017
Stb. non-forest	0.017	0.017	0.6030	0.0345	0	0
Stb. forest	0	0	0	0.2679	0	0.0168
Gain 2020	0	0.0029	0	0.0043	0.0043	0.0029
Gain 2010	0	0	0.0007	0.0002	0	0.0013

The change map achieved an overall accuracy of **0.8916 ± 0.0731**. However, this value is strongly influenced by a large proportion of stable classes. stable non-forest and stable forest together account for the largest share of the mapped area, meaning that high agreement in these classes has a strong effect on the overall accuracy. The change classes, especially forest gain and forest loss, showed much higher uncertainty and lower class-specific accuracies. Class-specific accuracy metrics were derived from the area-weighted confusion matrix.

Table 6: Accuracy metrics

	P.A	**UA**
Loss 2010	0.2095 ± 0.3164	0.5 ± 0.3267
Loss 2020	0.3219 ± 0.3705	0.6 ± 0.3201
Stb. non-forest	0.9928 ± 0.007	0.8974 ± 0.0965
Stb. forest	0.8625 ± 0.1327	0.9412 ± 0.1153
Gain 2020	1 ± 0	0.3 ± 0.2994
Gain 2010	0.0591 ± 0.0915	0.6 ± 0.3201

Class-specific accuracy values show that the stable classes were mapped most reliably. Stable forest reached a user's accuracy of **0.9412 ± 0.1153** and a producer's accuracy of **0.8625 ± 0.1327,** while the stable non-forest reached a user's accuracy of **0.8974 ± 0.0965** and a producer's accuracy of **0.9928 ± 0.007**. In contrast, forest loss and gain classes showed lower and more uncertain accuracies. Loss in 2010 had a particularlöy low producer's accuracy of **0.2095 ± 0.3164**, indicating that some reference loss areas were omitted from the mapped class. Gain in 2020 showed a low user's accuracy of **0.3 ± 0.2994,** suggesting substantial error. Although the producer's accuracy was estimated as **1 ± 0**, this value should be interpreted cautiously because of the limited number of reference samples for this class. This finally suggests that forest gain was more difficult to detect reliably than the stable classes. Overall, the results indicate that stable forest and stable non-forest were mapped more reliably than forest gain and forest loss.

4. Discussion

Land-cover and forest dynamics in Luxembourg show noticeable changes over the period 2000 to 2020, reflecting a gradual shift towards more human-dominated land-use patterns. Overall, the findings demonstrate a moderate but evident intensification of land use, primarily due to urban growth and a minor decrease in agricultural and forest areas. The growth in built-up areas in Luxembourg points to continued socio-economic progress and the expansion of infrastructure. While the total increase may seem relatively modest, even small additions of artificial surfaces can have a noticeable impact in a compact and highly interconnected country like Luxembourg, particularly in terms of landscape structure and ecological connectivity.

Forest cover remained one of the dominant land-use types over the study period, but overall, the results show a net loss of forest area. Based on the mapped data, forest decline is estimated at approximately 3321 hectares, whereas the area-adjusted figures indicate a much larger reduction of roughly 7281.04 hectares between 2000 and 2020. This difference between the two estimates highlights the importance of accounting for classification accuracy in change detection analysis. Because area-adjusted estimates correct for classification errors, they tend to provide a more realistic picture of actual land-cover change (Olofsson et al., 2014). In contrast, relying only on raw mapped pixel counts may lead to an underestimation of the true extent of forest loss and disturbance.

Forest loss in Luxembourg was unevenly distributed, occurring mainly at the edges of forest patches and in fragmented areas rather than within large continuous forests, likely due to local pressures such as logging, infrastructure expansion and urban growth. In addition, forest loss can be associated with broader socio-economic drivers, including increasing population density, housing demand and cross-border commuting pressures. Economic development and infrastructure investment have contributed to land conversion, particularly in peri-urban zones where accessibility and transport connectivity encourage urban sprawl. Agricultural intensification and changes in land management practices may also have indirectly influenced forest fragmentation in some regions.

Stable forest cover was mainly concentrated in northern Luxembourg (Oesling), while southern and central regions showed more stable non-forest land uses driven by urbanisation and agriculture. Forest gains were relatively small and scattered, suggesting that natural regrowth or afforestation is not keeping pace with forest loss, and these changes have important environmental implications. The reduction and fragmentation of forested areas can negatively affect biodiversity by isolating habitats and reducing ecological connectivity. In addition, forest loss contributes to decreased carbon storage capacity, which may weaken climate regulation functions. Other consequences include increased soil erosion risk, disruption of ecosystem services and overall landscape fragmentation, particularly in regions undergoing rapid urban expansion.

This points to a gradual shift from forested areas toward more human-dominated land uses, a pattern also observed in other parts of Central Europe where overall forest cover can appear stable despite underlying structural change. The accuracy assessment showed that stable land-cover classes were classified reliably, while change classes such as forest gain and loss were less certain. This is common in remote sensing studies, as change detection is more prone to classification errors and spectral confusion between transitional classes. Even though the approach is solid, there are still a few limitations to keep in mind. The 30 m resolution of Landsat imagery can miss finer details of land-cover change, particularly in areas where different land types are closely mixed, such as forest–urban transition zones. Similar vegetation types like forests, shrubs and transitional cover can also be difficult to distinguish, which may reduce classification accuracy.

The relatively limited number of validation samples may also introduce uncertainty into the accuracy assessment and the area-adjusted results. In addition, differences in image acquisition timing, cloud cover and atmospheric conditions can affect the consistency of the data. Future work could enhance this study by integrating higher resolution imagery, such as Sentinel-2 data, to improve the detection of fine-scale landcover changes. Extending the temporal scope beyond 2020 would also enable a more updated assessment of urban expansion and landscape dynamics. Moreover, incorporating seasonal classification strategies and advanced machine learning algorithms could strengthen the ability to distinguish between spectrally similar landcover classes.

Future research could also examine the links between land use change and climate variability while integrating socioeconomic data to provide deeper insight into the drivers of observed spatial changes. In summary, this study demonstrates the effectiveness of combining Landsat time-series data with a multi-temporal change analysis framework, in which land-cover transitions are tracked across multiple years alongside area-adjusted accuracy assessment (Potapov et al., 2022; Zhu et al., 2022). This approach provides a robust framework for understanding spatial and temporal forest dynamics and can be applied in other regions to support environmental monitoring, spatial planning and sustainable land management under conditions of ongoing land-use change.

5. Conclusion

This study highlights the value of integrating long-term Landsat-based datasets with a multi-temporal change detection approach for analysing landcover dynamics. By combining consistent Earth observation data with an accuracy-adjusted estimation method, it provides a reliable and structured way to quantify landcover transitions over extended time periods. The strength of this approach lies in its ability to support transparent and reproducible monitoring of environmental change using satellite image time series. It is particularly useful in regions where land use change occurs gradually, as it can detect subtle transitions that may not be evident in single date comparisons but still have important cumulative impacts on landscape structure and ecosystem processes. However, several challenges should also be considered. Remote sensing-based change detection is affected by classification uncertainties, spectral confusion between similar land-cover types and mixed pixels, which can reduce the reliability of change classes. In addition, limited reference data for rare transitions can lead to higher uncertainty in accuracy assessment. Temporal inconsistencies in satellite observations, atmospheric effects and differences in class definitions can also influence results and introduce bias if not properly addressed. Overall, the study demonstrates that multi-year remote sensing analysis combined with statistically sound accuracy assessment offers a robust basis for understanding long-term landcover dynamics. This method can be applied to support land management, spatial planning and environmental monitoring in other regions.

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Appendices

Appendix 1: GGE script for GLCLUC land-cover extraction of Luxembourg

```
var years = [2000, 2010, 2020];
var exportFolder = 'GEE_GLCLU_Luxembourg';
var exportScale = 30;  // GLCLU is 30 m Landsat-based data.
// GLAD GLCLU v2 assets.
var glcluAssets = {
  2000: 'projects/glad/GLCLU2020/v2/LCLUC_2000',
  2010: 'projects/glad/GLCLU2020/v2/LCLUC_2010',
  2020: 'projects/glad/GLCLU2020/v2/LCLUC_2020'
};
// GLAD ocean mask: keep land pixels only.
var landMask = ee.Image('projects/glad/OceanMask').lte(1);
//  Luxembourg boundary
var luxembourg = ee.FeatureCollection('FAO/GAUL/2015/level0')
  .filter(ee.Filter.eq('ADM0_NAME', 'Luxembourg'));
var luxGeom = luxembourg.geometry();
Map.centerObject(luxembourg, 9);
Map.addLayer(
  luxembourg.style({
    color: '000000',
    fillColor: '00000000',
    width: 2
  }),
  {},
  'Luxembourg boundary'
);
// Function to load and clip GLCLU maps
function loadGLCLU(year) {
  var img = ee.Image(glcluAssets[year])
    .updateMask(landMask)
    .clip(luxGeom)
    .rename('GLCLU_' + year)
    .toByte()
    .set({
      'year': year,
      'source': 'GLAD GLCLU / GLCLUC v2',
```

```
      'territory': 'Luxembourg'
    });
  return img;
}
// Create an ImageCollection for convenience.
var glcluCollection = ee.ImageCollection.fromImages(
  years.map(function(year) {
    return loadGLCLU(year);
  })
);
print('GLCLU Luxembourg ImageCollection', glcluCollection);
// Display palette is only for quick visualisation.
var displayVis = {
  min: 0,
  max: 255,
  palette: [
    'fefecd', // light land / sparse classes
    'b3b30e',
    '0d570d', // forest-like greens
    '643700',
    'bb00bb',
    '0986f7', // water-like blues
    '285d66',
    '991667',
    '0000ff',
    'ffffff'
  ]
};

years.forEach(function(year) {
  var img = loadGLCLU(year);
  Map.addLayer(img, displayVis, 'GLCLU Luxembourg ' + year, year === 2020);
});
// Optional: create a 3-band stacked image.
var glcluStack = ee.Image.cat([
  loadGLCLU(2000),
  loadGLCLU(2010),
  loadGLCLU(2020)
]);
```

```
print('Stacked 3-band GLCLU image', glcluStack);
// Optional class-code histograms
years.forEach(function(year) {
  var img = loadGLCLU(year);
  var hist = img.reduceRegion({
    reducer: ee.Reducer.frequencyHistogram(),
    geometry: luxGeom,
    scale: exportScale,
    maxPixels: 1e13,
    tileScale: 4
  });
  print('Class-code frequency histogram for Luxembourg, ' + year, hist);
});
// Export one GeoTIFF per year in Google Drive
years.forEach(function(year) {
  var img = loadGLCLU(year);
  Export.image.toDrive({
    image: img,
    description: 'GLCLU_Luxembourg_' + year,
    folder: exportFolder,
    fileNamePrefix: 'GLCLU_Luxembourg_' + year,
    region: luxGeom,
    scale: exportScale,
    maxPixels: 1e13,
    fileFormat: 'GeoTIFF',
    formatOptions: {
      cloudOptimized: true
    }
  });
});
// Optional export: stacked 2000/2010/2020 GeoTIFF
Export.image.toDrive({
  image: glcluStack,
  description: 'GLCLU_Luxembourg_2000_2010_2020_stack',
  folder: exportFolder,
  fileNamePrefix: 'GLCLU_Luxembourg_2000_2010_2020_stack',
  region: luxGeom,
  scale: exportScale,
  maxPixels: 1e13,
```

```
  fileFormat: 'GeoTIFF',
  formatOptions: {
    cloudOptimized: true
  }
});
```

Appendix 2: Rscript for Accuracy assessment

```
# ==============================================================
# AREA-ADJUSTED ACCURACY ASSESSMENT + AREA ESTIMATION
# Olofsson et al. (2014) style
# Using:
#   1) validation CSV with columns: label_map, label_reference
#   2) mapped class areas (hectares) from the full change map
# ==============================================================
# install.packages(c("readr", "dplyr"))
library(readr)
library(dplyr)
# ------------------------------------------------------------
# 1) FILE PATHS
# ------------------------------------------------------------
validation_csv_path <- "C:/Users/nkare/Desktop/RS. global change/validation_points2.csv"
output_file_main <- "C:/Users/nkare/Desktop/RS. global change/Accuracy_area_adjusted.csv"
output_file_reference <- "C:/Users/nkare/Desktop/RS. global change/Accuracy_area_adjusted_reference_summary.csv"
# ------------------------------------------------------------
# 2) CLASS DEFINITIONS
# ------------------------------------------------------------
valid_class_codes <- c("100", "110", "-1", "111", "001", "011")
class_names <- c(
  "100" = "Loss 2010",
  "110" = "Loss 2020",
  "-1"   = "Stable non-forest",
  "111"    = "Stable forest",
  "001"  = "Gain 2020",
  "011"  = "Gain 2010")
# ------------------------------------------------------------
# 3) MAPPED AREA BY CLASS (HECTARES)
# IMPORTANT: use POSITIVE area values
# ------------------------------------------------------------

mapped_area_hectares_by_class <- c(
  "100" = 2597.741022,
  "110" = 4333.422403,
  "-1"   = 174245.4669,
  "111"    = 73802.03843,
  "001"  = 3750.850244,
  "011"  = 577.9473187
)
total_roi_area_hectares <- sum(mapped_area_hectares_by_class)
print (total_roi_area_hectares)
if (total_roi_area_hectares <= 0) {
  stop("Total mapped area must be > 0.")
}
```

```
mapped_area_proportion_by_class <- mapped_area_hectares_by_class / total_roi_area_hectares
print(mapped_area_proportion_by_class)
# 95% confidence multiplier
confidence_multiplier <- 1.96
# ------------------------------------------------------------
# 4) READ VALIDATION CSV
# ------------------------------------------------------------
validation_data <- read_csv(validation_csv_path, show_col_types = FALSE)
validation_data <- validation_data %>%
  select(label_map, label_reference) %>%
  filter(!is.na(label_map), !is.na(label_reference))
validation_data$label_map <- as.numeric(validation_data$label_map)
validation_data$label_reference <- as.numeric(validation_data$label_reference)
validation_data <- validation_data %>%
  filter(label_map %in% valid_class_codes,
         label_reference %in% valid_class_codes)
if (nrow(validation_data) == 0) {
  stop("No valid observations found in the CSV after filtering.")
}
# ------------------------------------------------------------
# 5) SAMPLE CONFUSION MATRIX (COUNTS)
#    Rows = MAP
#    Columns = REFERENCE
# ------------------------------------------------------------
map_class_at_sample <- factor(validation_data$label_map, levels = valid_class_codes)
reference_class_at_sample <- factor(validation_data$label_reference, levels = valid_class_codes)
sample_confusion_counts <- table(map_class_at_sample, reference_class_at_sample)
sample_confusion_counts <- as.matrix(sample_confusion_counts)
cat("\nSample confusion matrix (raw counts):\n")
print(sample_confusion_counts)
# ------------------------------------------------------------
# 6) AREA-PROPORTION ERROR MATRIX
# ------------------------------------------------------------
sample_total_per_map_class <- rowSums(sample_confusion_counts)
print(sample_total_per_map_class)
# protect against division by zero
sample_proportion_matrix <- sweep(sample_confusion_counts, 1, sample_total_per_map_class, "/")
sample_proportion_matrix[is.nan(sample_proportion_matrix)] <- 0
print(sample_proportion_matrix)
estimated_area_proportion_matrix <- sweep(
  sample_proportion_matrix,
  1,
  mapped_area_proportion_by_class[rownames(sample_confusion_counts)],
  "*"
)
print(estimated_area_proportion_matrix)
estimated_area_proportion_matrix[is.nan(estimated_area_proportion_matrix)] <- 0
# ------------------------------------------------------------
# 7) ACCURACY METRICS
# ------------------------------------------------------------
overall_accuracy <- sum(diag(estimated_area_proportion_matrix))
users_accuracy <- diag(estimated_area_proportion_matrix) / rowSums(estimated_area_proportion_matrix)
producers_accuracy <- diag(estimated_area_proportion_matrix) / colSums(estimated_area_proportion_matrix)
users_accuracy[is.nan(users_accuracy)] <- NA
producers_accuracy[is.nan(producers_accuracy)] <- NA
# ------------------------------------------------------------
# 8) UNCERTAINTY
# ------------------------------------------------------------
```

```
standard_error_overall_accuracy <- sqrt(
  sum(
    (mapped_area_proportion_by_class[rownames(sample_confusion_counts)]^2) *
      users_accuracy * (1 - users_accuracy) /
      (sample_total_per_map_class - 1),
    na.rm = TRUE
  )
)
confidence_half_width_overall_accuracy <- confidence_multiplier * standard_error_overall_accuracy
standard_error_users_accuracy <- sqrt(
  users_accuracy * (1 - users_accuracy) / (sample_total_per_map_class - 1)
)
confidence_half_width_users_accuracy <- confidence_multiplier * standard_error_users_accuracy
# Producer's accuracy uncertainty
estimated_reference_area_total <- rep(NA_real_, length(valid_class_codes))
names(estimated_reference_area_total) <- as.character(valid_class_codes)
second_term_sum <- rep(0, length(valid_class_codes))
names(second_term_sum) <- as.character(valid_class_codes)
for (reference_class_code in valid_class_codes) {
  reference_class_code_text <- as.character(reference_class_code)
  estimated_reference_area_total[reference_class_code_text] <- sum(
    (mapped_area_hectares_by_class / sample_total_per_map_class) *
      sample_confusion_counts[, reference_class_code_text],
    na.rm = TRUE
  )
  other_map_classes <- setdiff(valid_class_codes, reference_class_code)
  if (length(other_map_classes) > 0) {
    contribution <- (mapped_area_hectares_by_class[as.character(other_map_classes)]^2) *
      (sample_confusion_counts[as.character(other_map_classes), reference_class_code_text] /
         sample_total_per_map_class[as.character(other_map_classes)]) *
      (1 - (sample_confusion_counts[as.character(other_map_classes), reference_class_code_text] /
              sample_total_per_map_class[as.character(other_map_classes)])) /
      (sample_total_per_map_class[as.character(other_map_classes)] - 1)

    second_term_sum[reference_class_code_text] <- sum(contribution, na.rm = TRUE)
  }
}
standard_error_producers_accuracy <- rep(NA_real_, length(valid_class_codes))
names(standard_error_producers_accuracy) <- as.character(valid_class_codes)
for (reference_class_code in valid_class_codes) {
  reference_class_code_text <- as.character(reference_class_code)
  if (is.na(producers_accuracy[reference_class_code_text]) ||
      is.na(users_accuracy[reference_class_code_text]) ||
      is.na(estimated_reference_area_total[reference_class_code_text]) ||
      estimated_reference_area_total[reference_class_code_text] == 0) {
    standard_error_producers_accuracy[reference_class_code_text] <- NA
    next
  }
  first_term <- (mapped_area_hectares_by_class[reference_class_code_text]^2) *
    (1 - producers_accuracy[reference_class_code_text])^2 *
    (users_accuracy[reference_class_code_text] * (1 - users_accuracy[reference_class_code_text]) /
       (sample_total_per_map_class[reference_class_code_text] - 1))
  second_term <- (producers_accuracy[reference_class_code_text]^2) *
    second_term_sum[reference_class_code_text]
  variance_producers_accuracy <- (first_term + second_term) /
    (estimated_reference_area_total[reference_class_code_text]^2)
  standard_error_producers_accuracy[reference_class_code_text] <- sqrt(variance_producers_accuracy)
}
```

```
confidence_half_width_producers_accuracy <- confidence_multiplier * standard_error_producers_accuracy
# ------------------------------------------------------------
# 9) REFERENCE-ADJUSTED AREA ESTIMATION
# ------------------------------------------------------------
adjusted_area_proportion_by_reference_class <- colSums(estimated_area_proportion_matrix)
adjusted_area_hectares_by_reference_class <- adjusted_area_proportion_by_reference_class *
total_roi_area_hectares
standard_error_adjusted_area_proportion <- rep(NA_real_, length(valid_class_codes))
names(standard_error_adjusted_area_proportion) <- as.character(valid_class_codes)
for (reference_class_code in valid_class_codes) {
  reference_class_code_text <- as.character(reference_class_code)
  sample_fraction_in_reference_class <- sample_confusion_counts[, reference_class_code_text] /
    sample_total_per_map_class
  standard_error_adjusted_area_proportion[reference_class_code_text] <- sqrt(
    sum(
      mapped_area_proportion_by_class[rownames(sample_confusion_counts)]^2 *
        sample_fraction_in_reference_class *
        (1 - sample_fraction_in_reference_class) /
        (sample_total_per_map_class - 1),
      na.rm = TRUE
    )
  )
}
confidence_half_width_adjusted_area_proportion <- confidence_multiplier *
standard_error_adjusted_area_proportion
confidence_half_width_adjusted_area_hectares <- confidence_half_width_adjusted_area_proportion *
total_roi_area_hectares
# ------------------------------------------------------------
# 10) BUILD OUTPUT TABLES
# ------------------------------------------------------------
main_table <- as.data.frame.matrix(round(estimated_area_proportion_matrix, 6))
colnames(main_table) <- unname(class_names[colnames(main_table)])
rownames(main_table) <- unname(class_names[rownames(main_table)])
main_table$`Sample size in mapped class` <-
sample_total_per_map_class[rownames(sample_confusion_counts)]
main_table$`Mapped area in ROI (ha)` <-
round(mapped_area_hectares_by_class[rownames(sample_confusion_counts)], 2)
main_table$`Mapped area proportion in ROI` <-
round(mapped_area_proportion_by_class[rownames(sample_confusion_counts)], 6)
main_table$`User's accuracy` <- round(users_accuracy[rownames(sample_confusion_counts)], 4)
main_table$`95% CI half-width for user's accuracy` <-
round(confidence_half_width_users_accuracy[rownames(sample_confusion_counts)], 4)
main_table$`Overall accuracy` <- NA_real_
main_table$`95% CI half-width for overall accuracy` <- NA_real_
main_table$`Overall accuracy`[1] <- round(overall_accuracy, 4)
main_table$`95% CI half-width for overall accuracy`[1] <- round(confidence_half_width_overall_accuracy, 4)
reference_summary_table <- data.frame(
  `Reference class code` = valid_class_codes,
  `Reference class name` = unname(class_names[as.character(valid_class_codes)]),
  `Adjusted proportion of area` =
round(adjusted_area_proportion_by_reference_class[as.character(valid_class_codes)], 6),
  `95% CI half-width (proportion)` =
round(confidence_half_width_adjusted_area_proportion[as.character(valid_class_codes)], 6),
  `Adjusted area (hectares)` =
round(adjusted_area_hectares_by_reference_class[as.character(valid_class_codes)], 2),
  `95% CI half-width (hectares)` =
round(confidence_half_width_adjusted_area_hectares[as.character(valid_class_codes)], 2),
  `Producer's accuracy` = round(producers_accuracy[as.character(valid_class_codes)], 4),
```

```
    `95% CI half-width for producer's accuracy` = round(confidence_half_width_producers_accuracy[as.character(valid_class_codes)], 4)
)
# ------------------------------------------------------------
# 11) WRITE OUTPUT FILES
# ------------------------------------------------------------
write.csv(main_table, output_file_main, row.names = TRUE, quote = FALSE)
write.csv(reference_summary_table, output_file_reference, row.names = FALSE, quote = FALSE)
```

Appendix 3: Class Statistics

	2000	**2010**	**2020**
classes	**Area (ha)**	**Area (ha)**	**Area (ha)**
semi-arid	123.76	63.20	68.82
Dense short vegetation	54192.15	50702.99	50219.56
open forest	11852.48	11326.50	10063.42
dense forest	69553.47	67441.73	68021.24
wetland salt pan	6.77	0.85	1.80
wetland spare vegetation	15.48	1.38	0.90
wetland dense short vegetation	81.53	57.97	45.04
wetland Tree cover	56.79	62.51	58.76
Open surface water	641.56	581.46	589.08
cropland	40478.21	38802.72	38359.96
Built-up	82297.12	90263.97	91873.62